

RUI XING

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TECHNICAL SKILLS

Core Domain	Large-Scale Optimization, GenAI Agents, and AWS ML Infrastructure.
Language	Java, Python, C, C++, Typescript, SQL, Bash
Technologies	AWS (Bedrock, ECS, Lambda, Stepfunction, DynamoDB), CDK, Docker

RELEVANT WORK EXPERIENCE

Amazon	Bellevue, WA
<i>Software Development Engineer</i>	2020 – Present

Large-Scale Order Quantity Optimization Engine 2025

- Architected and delivered an end-to-end Order Quantity Optimization system for Amazon supply chain planning, using Column Generation decomposition to scale million-variable models and reduce production model-build time.
- Developed the core high-performance engine in Java, implementing custom data structures and a selective variable pricing mechanism that minimized memory overhead and reduced convergence runtime by 9.5%.
- Optimized system throughput by implementing multi-threaded execution for candidate evaluation, achieving a 20% reduction in end-to-end runtime and maximizing hardware utilization during high-load scenarios.
- Implemented prototype variable bucketing and Column Generation toggles; evaluated lightweight models (Decision Trees, Linear Regression) to guide toggling decisions and explore runtime–convergence trade-offs.

Complex Workflow LLM Agent/GenAI Observability 2025

- Initiated, built, and demoed a GenAI observability MVP for massive-scale distributed simulations, comparing near-real-time run velocity to historical baselines to deliver one-click anomaly detection for on-call triage.
- Implemented the workflow with AWS Bedrock agentic orchestration and Lambda-backed tool APIs to query heterogeneous sources and generate evidence-linked investigation narratives.
- Drove the initiative forward by working with a sister team after the demo to refine the workflow and provide code/design walkthroughs, supporting their productionization effort.

Simulation Fidelity & Sensitivity Analysis 2024

- Developed a sensitivity-analysis framework to de-risk an upstream dependency deprecation, running high-concurrency dataset perturbations and parallel service calls to quantify data-fidelity gaps and guide rollout decisions.

Data Modeling 2023

- Re-architected a production DynamoDB data model with an access-pattern-based key design to preserve historical state, eliminate update-driven data loss, and improve auditability.
- Designed a backward-compatible, zero-downtime migration using an idempotent one-time backfill plus stream-driven change propagation for live updates.

MLE Execution Platform 2022

- Built the 0→1 AWS IaC (Infrastructure as Code) foundation for a production optimization service, standardizing multi-environment deployments and release controls through a reproducible framework.
- Engineered a greenfield event-driven orchestration engine (EventBridge, Lambda, ECS) for containerized scientist-authored Python workflows, enabling parallel task runs and reducing end-to-end runtime by 33%.

EDUCATION

Bachelor of Computer Science, Honours, Co-op	January 2016 - June 2020
University of Waterloo, Canada	